

VexarDrive Fleet Analytics & Risk Intelligence

Technical Report

Project: Data Scientist Intern Assignment

Organization: VexarDrive Technologies

Tools: Python, Pandas, NumPy, Scikit-learn, Power BI

Dataset: One week of two-wheeler delivery/rider fleet telemetry

1. Executive Summary

This project develops an end-to-end fleet analytics solution using one week of operational and telemetry data from VexarDrive Technologies.

The dataset contains **30 drivers, 30 vehicles, 450 trips**, and per-minute GPS, accelerometer, and gyroscope readings captured during trips. The assignment requires two analytical dashboards: a **Driver Behaviour Dashboard** and a **Vehicle Health Status Dashboard**.

The solution was developed in two major analytical streams:

1. Driver Behaviour & Risk Analytics

- Identifies risky driving behaviours.
- Aggregates risk events at driver level.
- Calculates a composite 0–100 Driver Risk Score.
- Classifies drivers into High, Moderate, and Low Risk categories.

2. Vehicle Health & Sensor Anomaly Analytics

- Derives accelerometer and gyroscope variability measures.
- Detects unusual sensor spikes using fleet-relative thresholds.
- Calculates a 0–100 Vehicle Health Score.
- Identifies vehicles requiring maintenance review.

The final results identify **8 high-risk drivers, 14 moderate-risk drivers, and 8 low-risk drivers**. The vehicle analysis identifies **15 maintenance-review candidates**, 10 vehicles to monitor, and 5 vehicles classified as relatively healthy.

The solution is intentionally interpretable: every composite score is derived from identifiable behavioural or sensor features rather than being treated as a black-box prediction.

2. Business Problem

VexarDrive operates a connected two-wheeler delivery fleet. The available data combines:

- Driver master information
- Vehicle master information
- Trip summaries
- GPS telemetry
- Accelerometer telemetry
- Gyroscope telemetry

The dataset structure specifies one row per driver, one row per vehicle, one row per trip, and one telemetry observation per minute per trip.

The business problem is therefore divided into two questions:

Driver question

Which drivers demonstrate risky driving behaviour, and what behaviours contribute most to their risk?

Vehicle question

Which vehicles exhibit unusual sensor behaviour that may warrant maintenance review?

The assignment specifically requires the scores and dashboard metrics to be supported by reasoning, methodology, and assumptions.

3. Project Objectives

Objective 1 — Driver Behaviour Analytics

Develop a driver-level risk framework that:

- Detects risky driving events.
- Aggregates events by driver.
- Measures event frequency.
- Identifies the dominant risk factor.
- Produces a normalized Driver Risk Score.
- Categorizes drivers according to risk level.

Objective 2 — Vehicle Health Analytics

Develop a vehicle-level sensor-health framework that:

- Measures acceleration variability.
- Measures gyroscope variability.
- Detects sensor spikes.
- Aggregates telemetry behaviour by vehicle.
- Produces a Vehicle Health Score.
- Prioritizes vehicles for maintenance review.

Objective 3 — Management Visualization

Build two interactive Power BI dashboards:

- Driver Behaviour Dashboard
- Vehicle Health Status Dashboard

Objective 4 — Extensibility

Identify additional analytical use cases that can be developed from the same telemetry infrastructure.

4. Dataset Overview

The assignment dataset contains:

Entity	Records
Drivers	30
Vehicles	30
Trips	450
Trips per Driver	15
Telemetry	Per-minute trip readings
Time Period	1 week

The official data relationships are:

Telemetry.Trip_ID
↓
Trips.Trip_ID

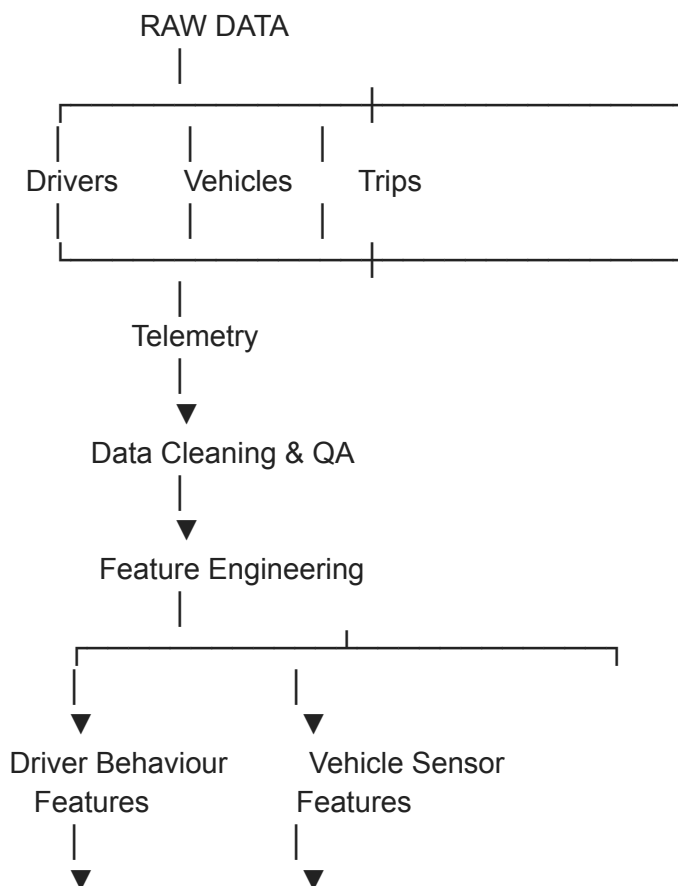
Trips.Driver_ID
↓
Drivers.Driver_ID

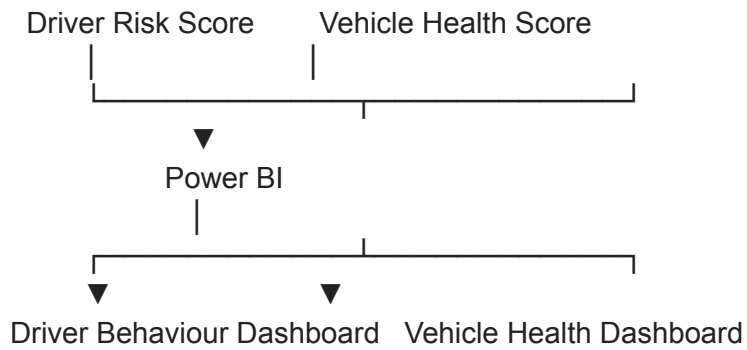
Trips.Vehicle_ID
↓
Vehicles.Vehicle_ID

These relationships are explicitly defined in the assignment brief.

5. Analytical Architecture

The overall solution follows this architecture:





6. Data Preparation

6.1 Data Cleaning

The data preparation stage includes:

- Data-type validation
- Missing-value analysis
- Duplicate detection
- ID validation
- Timestamp conversion
- Chronological ordering
- Sensor-value validation
- Relationship validation

Telemetry is ordered chronologically within each trip before sequential calculations are performed.

6.2 Data Granularity

The project deliberately preserves three analytical levels:

Driver level

Used for:

- Risk scoring
- Driver ranking
- Behaviour comparison

Trip level

Used for:

- Event detection
- Trip-level behavioural analysis
- Sensor variability

Telemetry level

Used for:

- Sensor calculations
- Spike detection
- Sequential changes
- GPS/IMU analysis

This avoids prematurely aggregating the raw telemetry and losing useful information.

7. Driver Behaviour Analytics

7.1 Behavioural Events

The driver analysis considers several risky behaviours:

- Harsh acceleration
- Harsh braking
- Sudden turning
- High-speed events
- Overall high-risk-event frequency

Each event is detected at trip/telemetry level and subsequently aggregated to driver level.

7.2 Driver-Level Features

The primary driver-level metrics are:

Total_Risk_Events

Risk_Events_Per_Trip

Primary_Risk_Factor

Total Risk Events

Measures the total number of detected risky events attributed to the driver.

Risk Events per Trip

Normalizes event count by the number of trips:

$$\text{RiskEventsPerTrip} = \frac{\text{TotalRiskEvents}}{\text{NumberOfTrips}}$$

This prevents drivers with more recorded trips from appearing riskier simply because they have more observations.

In this dataset, each driver has 15 trips, but normalization still makes the metric more generalizable.

8. Driver Risk Score

A composite Driver Risk Score was developed on a **0–100 scale**.

Interpretation:

0 → lowest relative risk

100 → highest relative risk

The score combines multiple behavioural indicators rather than relying on a single event type.

This produces an interpretable ranking of drivers.

9. Driver Risk Results

The final driver analysis produced:

Risk Category	Driver s
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High Risk	8
Moderate Risk	14
Low Risk	8
Total	30

Score statistics

Statistic	Value
Mean	51.67
Minimum	6.17
Maximum	98.42

9.1 Highest-Risk Driver

D24 — Lakshmi Iyer

Risk Score: 98.42

Primary risk factor:

Harsh braking

Additional results:

- Total risk events: **220**
- Risk events per trip: **14.67**
- Risk category: **High Risk**

This makes D24 the highest-priority candidate for behavioural intervention.

9.2 Other High-Risk Drivers

Rank	Driver	Score	Primary Factor
1	Lakshmi Iyer	98.42	Harsh braking
2	Bhavani Raj	95.17	Harsh acceleration
3	Kavya Pillai	93.08	Harsh acceleration
4	Rajesh Subramaniam	91.00	High risk-event frequency
5	Vignesh Murugan	85.50	Harsh acceleration

This shows that high risk is not driven by a single behaviour across all drivers.

10. Vehicle Health Analytics

The second analytical stream uses accelerometer and gyroscope telemetry to identify unusual vehicle sensor signatures.

The assignment specifically asks for vehicles showing signs of mechanical wear or irregular sensor signatures that may need maintenance.

Because the dataset does **not** provide confirmed mechanical-failure labels, the analysis treats the problem as **relative sensor anomaly detection**, not supervised mechanical-failure prediction.

11. Accelerometer Feature Engineering

The accelerometer contains three axes:

Accel_X
Accel_Y
Accel_Z

The overall acceleration magnitude is calculated as:

$$\sqrt{\text{AccelMagnitude} = \sqrt{\text{Accel_X}^2 + \text{Accel_Y}^2 + \text{Accel_Z}^2}}$$

This creates a single measure of overall acceleration intensity independent of individual axis direction.

12. Gyroscope Feature Engineering

The gyroscope contains:

Gyro_X
Gyro_Y
Gyro_Z

The rotational magnitude is calculated as:

$$\sqrt{\text{Gyro_X}^2 + \text{Gyro_Y}^2 + \text{Gyro_Z}^2}$$

This provides a combined measure of rotational movement.

13. Vehicle-Level Sensor Features

The telemetry is aggregated to vehicle level using:

1. Average acceleration variability

Avg_Accel_Variability

Higher values indicate greater variation in acceleration behaviour.

2. Average gyroscope variability

Avg_Gyro_Variability

Higher values indicate greater variation in rotational behaviour.

3. Average sensor spike frequency

Avg_Total_Sensor_Spikes

Higher values indicate more frequent unusual sensor observations.

14. Sensor Spike Detection

Instead of selecting arbitrary engineering thresholds, the project uses fleet-relative percentile thresholds.

The previously calculated fleet percentiles provide reference points for:

- Speed changes
- Acceleration changes
- Gyroscope changes
- Speed

For example, the acceleration-change analysis showed:

50th percentile = 0.046

90th percentile = 0.146

95th percentile = 0.216

99th percentile = 0.588

The gyroscope-change distribution showed a much larger upper tail:

50th percentile = 1.42

90th percentile = 4.97

95th percentile = 32.99

99th percentile = 47.53

These distributions demonstrate why percentile-based detection is preferable to casually selecting one arbitrary threshold.

15. Vehicle Health Score

The final Vehicle Health Score combines three dimensions:

Component	Weight
Acceleration Health	40%

Gyroscope Health	35%
Sensor Spike Health	25%
Total	100%

The score is normalized to:

$$[0 \leq \text{VehicleHealthScore} \leq 100]$$

Interpretation:

100 = relatively healthy sensor behaviour

0 = relatively high sensor anomaly burden

The components are calculated using fleet-relative percentile rankings and then inverted so that higher values represent better health.

16. Vehicle Health Categories

Score	Classification
75–100	Healthy
50–74.9	Monitor
9	
<50	Maintenance Review

These are **analytical screening bands**.

They should not be interpreted as:

- OEM limits
 - Engineering failure thresholds
 - Safety certification limits
 - Confirmed mechanical-failure boundaries
-

17. Vehicle Health Results

The final analysis produced:

Category	Vehicle s
Maintenance Review	15
Monitor	10
Healthy	5
Total	30

Score statistics

Statisti c	Valu e
Mean	48.3 3
Median	50.9 2
Minimu m	3.33
Maximu m	92.8 3

18. Highest Maintenance-Review Candidates

V19

Health Score: 3.33

Primary concern:

High acceleration variability

Supporting metrics:

Avg Acceleration Variability = 0.120131
Avg Gyroscope Variability = 8.685968
Avg Sensor Spikes = 2.047166

V19 is particularly significant because its sensor profile is elevated across all three major health dimensions.

V02

Health Score: 19.83

Primary concern:

High acceleration variability

Supporting metrics:

Avg Acceleration Variability = 0.150929
Avg Gyroscope Variability = 6.074134
Avg Sensor Spikes = 2.047751

V02 also has an elevated sensor-spike profile.

V24

Health Score: 13.00

Primary concern:

High gyroscope variability

Supporting metrics:

Avg Acceleration Variability = 0.086439
Avg Gyroscope Variability = 8.788835
Avg Sensor Spikes = 1.482013

V24 demonstrates why multiple indicators are useful: its dominant concern is gyroscope variability rather than acceleration variability.

19. Power BI Data Model

The Power BI model uses master tables as dimensions and detailed tables as analytical/fact-style tables.

Recommended relationships:

Drivers Clean



Trip Behaviour Features

Vehicles Clean

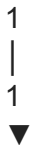


Vehicle Trip Features

Trip Behaviour Features

Scoring tables connect to their corresponding master entities:

Drivers Clean



Driver Risk Scores

Vehicles Clean





Vehicle Health Scores

Direct relationships between detailed analytical tables are avoided where possible to prevent ambiguous filter paths.

20. Dashboard Design

20.1 Driver Behaviour Dashboard

The dashboard contains:

KPI Cards

- Total Drivers
- Average Risk Score
- High-Risk Drivers
- Total Risk Events

Visuals

- Driver Risk Ranking
- Risk Category Distribution
- Primary Risk Factors
- Risk Events by Driver
- Driver Detail Table

Filters

- Driver Name
 - Risk Category
-

20.2 Vehicle Health Dashboard

The dashboard contains:

KPI Cards

- Total Vehicles
- Average Vehicle Health
- Maintenance Review
- High Priority Vehicles

Visuals

- Vehicle Health Ranking
- Health Category Distribution
- Primary Vehicle Health Concerns
- Maintenance Priority
- Vehicle Maintenance Detail Table

Filters

- Vehicle ID
 - Vehicle Type
 - Make
-

21. Detail Pages

The solution can be extended with individual detail pages.

Driver Detail

The Driver Detail page contains:

- Driver Risk Score
- Total Risk Events
- Risk Events per Trip
- Risk Category
- Primary Risk Factor
- Trip-level behaviour
- Behaviour trends

For example:

D24

Lakshmi Iyer

Risk Score 98.42

Risk Category High Risk

Risk Events 220
Events/Trip 14.67
Primary Factor Harsh braking

Vehicle Detail

The Vehicle Detail page contains:

- Vehicle Health Score
- Maintenance Priority
- Maintenance Rank
- Primary Health Concern
- Acceleration variability
- Gyroscope variability
- Sensor spike frequency
- Trip-level sensor trends

For V19:

Vehicle Health Score 3.33
Maintenance Rank 1
Priority High
Primary Concern High acceleration variability

22. Validation Strategy

The Power BI outputs are validated against the Python-generated results.

Driver validation

Expected:

Drivers = 30
High Risk = 8
Moderate Risk = 14
Low Risk = 8
Average Risk $\approx$ 51.67

Vehicle validation

Expected:

Vehicles = 30

Maintenance Review = 15

Monitor = 10

Healthy = 5

Average Health ≈ 48.33

The top-ranked entities are also manually checked.

This ensures that dashboard calculations have not inadvertently changed the analytical results generated in Python.

23. Assumptions

Assumption 1 — Relative scoring

Vehicle health is calculated relative to the observed fleet.

Therefore, scores may change when new vehicles or additional historical data are introduced.

Assumption 2 — No failure labels

There are no confirmed mechanical-failure labels in the provided dataset.

Therefore, the Vehicle Health Score is an anomaly/maintenance-review score.

Assumption 3 — Sensor anomalies are not necessarily faults

An unusual sensor reading may result from:

- Road conditions
- Legitimate manoeuvres
- Driver behaviour
- Sensor noise
- Data-quality issues

Therefore, maintenance review is recommended rather than automatic fault diagnosis.

Assumption 4 — Composite weights

The Vehicle Health Score uses:

40% acceleration
35% gyroscope
25% sensor spikes

These weights are analytical choices and should ideally be validated against historical maintenance outcomes.

24. Limitations

Limited observation window

The dataset covers only one week.

Therefore, the analysis cannot establish long-term driver or vehicle trends.

Lack of ground truth

There are no confirmed mechanical-failure labels.

Small fleet

Only 30 vehicles and 30 drivers are available.

Sensor-based inference

Sensor anomalies do not necessarily represent mechanical defects.

Threshold sensitivity

Percentile-based anomaly detection depends on the observed fleet distribution.

25. Recommended Business Actions

Driver Management

For high-risk drivers:

1. Review individual behaviour patterns.
2. Identify dominant risk factors.
3. Provide targeted coaching.
4. Monitor risk score after intervention.
5. Compare risk-event frequency over time.

For example:

D24 → focus on harsh braking.

Rather than providing generic driver training, intervention can target the specific behaviour contributing most strongly to the driver's risk profile.

Vehicle Maintenance

For high-priority vehicles:

1. Review telemetry history.
2. Check recent service records.
3. Inspect the relevant sensor patterns.
4. Perform physical inspection where appropriate.
5. Compare future telemetry after servicing.

For example:

V19 → prioritize review because multiple sensor indicators are elevated.

26. Future Improvements

26.1 Predictive Maintenance

If historical maintenance/failure labels become available, train supervised models such as:

- XGBoost
- LightGBM
- Random Forest

- Logistic Regression

The target could be:

Maintenance Required

0 = No

1 = Yes

This would transform the current anomaly-ranking approach into predictive maintenance.

26.2 Time-Series Anomaly Detection

Use:

- Isolation Forest
- Autoencoders
- One-Class SVM
- Change-point detection

to detect anomalous telemetry windows instead of relying primarily on aggregated trip statistics.

26.3 Longitudinal Risk Monitoring

With additional weeks/months of data:

Driver Risk

Week 1 → 42

Week 2 → 48

Week 3 → 61

Week 4 → 73

could reveal deteriorating driving behaviour.

Similarly:

Vehicle Health

Week 1 → 82

Week 2 → 74

Week 3 → 61

Week 4 → 45

could identify deteriorating vehicles.

27. Step 10 — Driver & Vehicle Detail Pages

The Power BI solution can include dedicated pages for:

Driver Detail

- Driver score
- Risk category
- Risk events
- Risk events per trip
- Primary risk factor
- Trip-level behaviour
- Historical trends

Vehicle Detail

- Health score
- Maintenance priority
- Primary sensor concern
- Sensor variability
- Sensor spikes
- Trip-level sensor trends

This allows users to move from fleet-level analysis to individual investigation.

28. Step 11 — Drill-Through Analysis

The next improvement is Power BI drill-through.

A fleet manager could select:

Driver Behaviour



D24



Driver Detail

and automatically see D24's complete risk profile.

Similarly:

Vehicle Health



V19



Vehicle Detail

would open V19's sensor-health profile.

This creates a hierarchical analytical experience:

Fleet Overview



Driver / Vehicle Ranking



Individual Entity



Detailed Behaviour / Sensor Analysis

29. Additional Analytical Opportunities

The assignment specifically asks what else the dataset could be used for beyond the two dashboards.

Potential future applications include:

Driver coaching optimization

Identify exactly which behaviour each driver should improve.

Route risk analysis

Compare risky events across routes or geographic regions.

Delivery efficiency

Analyze:

- Trip duration
- Distance
- Speed
- Event frequency

Fuel/energy efficiency

If fuel or battery information becomes available, connect driving behaviour with energy consumption.

Fleet utilization

Analyze vehicle usage and trip allocation.

Predictive maintenance

Combine telemetry with service history and failure labels.

Safety monitoring

Build near-real-time alerts for severe driving events.

30. Conclusion

The VexarDrive Fleet Analytics project converts raw driver, vehicle, trip, GPS and IMU telemetry into an interpretable fleet intelligence system.

The project successfully establishes two analytical frameworks:

Driver Risk

Behavioural Events



Driver Features



Risk Score



Risk Category



Targeted Coaching

Vehicle Health

Sensor Telemetry



Variability + Spike Features



Fleet-relative Anomaly Scores



Health Score



Maintenance Review Priority

The analysis identified:

- **8 high-risk drivers**
- **14 moderate-risk drivers**
- **8 low-risk drivers**
- **15 maintenance-review vehicles**
- **10 vehicles requiring monitoring**
- **5 relatively healthy vehicles**

The strongest driver-risk candidate is **D24 (Lakshmi Iyer)** with a score of **98.42**, primarily associated with harsh braking.

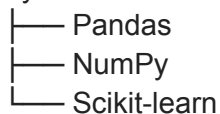
The strongest vehicle maintenance-review candidate is **V19**, with a health score of **3.33** and elevated acceleration, gyroscope, and sensor-spike indicators.

The resulting Power BI architecture provides both an executive-level fleet view and a path toward detailed driver/vehicle investigation.

Most importantly, the project distinguishes **analytical anomaly detection from confirmed mechanical diagnosis**, making the recommendations appropriately cautious given the available data.

Technical Stack

Python



Business Intelligence



Development

└─ Jupyter Notebook

Project Deliverables

- ✓ Data cleaning
- ✓ Exploratory analysis
- ✓ Driver behaviour feature engineering
- ✓ Driver Risk Score
- ✓ Vehicle sensor feature engineering
- ✓ Sensor anomaly detection
- ✓ Vehicle Health Score
- ✓ Maintenance prioritization
- ✓ Driver Behaviour Dashboard
- ✓ Vehicle Health Dashboard
- ✓ Detail-page architecture
- ✓ Drill-through architecture
- ✓ Business recommendations
- ✓ Assumptions and limitations